

SOFTWARE ARCHITECTURES INTRODUCTION

Software Engineering Seminar

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

.

1 Systems Thinking

2 Reference Architectures



Outline

1 Systems Thinking

2 Reference Architectures



What is a System?

- A **system** is a set of interacting components that work together to achieve a common goal.
 → modules - classes
- A **system** is a collection of elements that are organized in a specific way.
 ⇒ architecture
- A **system** is a structure that is designed to perform a specific function.
 →

A-B-C

A-B
O

B
A-C



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- **Systems analysis** is the *process* of **understanding** how a **system works** and how it can be **improved**.
- **Systems analysis** is the *first step* in the **systems development lifecycle**.

Car



LLM

```

    graph TD
      fire --- turning
  
```



Systems Analysis Techniques

- **Systems analysis** uses a variety of techniques to study a system.
- It includes interviews, surveys, observations, and document analysis.
- It also includes data modeling, process modeling, and requirements analysis.

final users

stakeholders

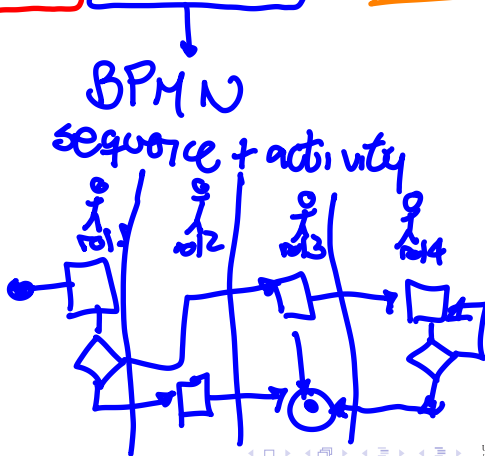
business processes

flowchart

diagrams



- **Systems analysis** uses a variety of techniques to study a system.
- It includes interviews, surveys, observations, and document analysis.
- It also includes data modeling, process modeling, and requirements analysis.



- Sources
- Format + structure
- Understanding



Complexity and Emergence

- **Complexity** is the *degree* to which a **system** is difficult to understand.

- Emergence is the *appearance of unexpected properties* in a system

- They form from interactions of its components
- Complexity and emergence are common in dynamic systems that are non-linear.

- They can be studied and understood through systems analysis and modeling.

+ elements
+ relations

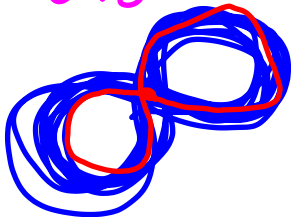
chaos
theory

randomness
(stochastic)

non-linear

chaotic
attractors

Lorenz

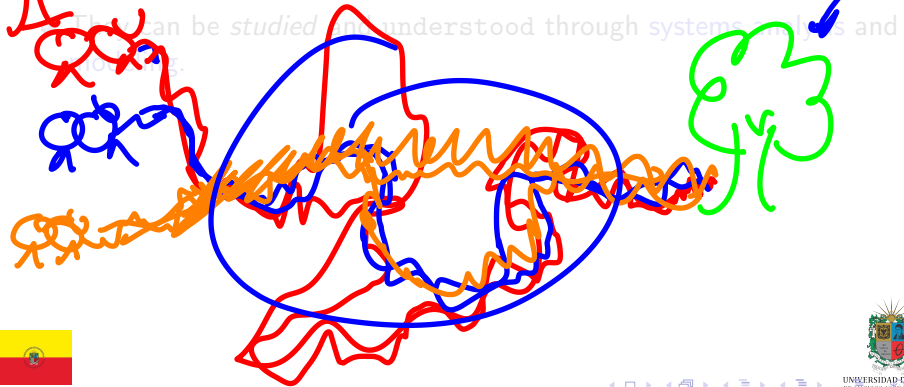


Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.
- **Emergence** is the appearance of unexpected properties in a system that arise from the interactions of its components.

- Complexity and emergence are common in dynamic systems that are nonlinear and chaotic.

swarm intelligence



Complexity and Emergence

- **Complexity** is the *degree* to which a **system** is **difficult** to **understand**.
- **Emergence** is the *appearance* of **unexpected properties** in a **system** that **arise** from the **interactions** of its **components**.
- **Complexity** and **emergence** are common in **dynamic systems** that are **non-linear** and **chaotic**.
- They can be *studied* and understood through **systems analysis** and **modeling**.



Strategies to Solve Problems

- **Top-Down:** Start from the big picture and break it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often *combined* in software design.



Strategies to Solve Problems

- **Top-Down:** Start from the big picture and break it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often combined in software design.



Strategies to Solve Problems

- **Top-Down:** Start from the big picture and break it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often *combined* in software design.



Process Definition

- A **Process** is a **series** of steps or actions taken to achieve a particular end.
- **Processes** are used to **organize** and **manage** work.



Workflows

- A **Workflow** is a **series** of tasks that are performed in a specific order to *achieve a goal*.
- **Workflows** are used to *automate* and *optimize* **business processes**.
- **Workflows** can be **sequential**, **parallel**, **conditional**, or **repetitive**.



Process Models

- A **Process Model** is a representation of a **process** that shows the sequence of steps and the **relationships** between them.
- **Process models** are used to **analyze**, **design**, and **improve** processes.
- Examples of **process models** include flowcharts, data flow diagrams, activity diagrams, business process model and notation (BPMN), petri nets, state diagrams, among others.

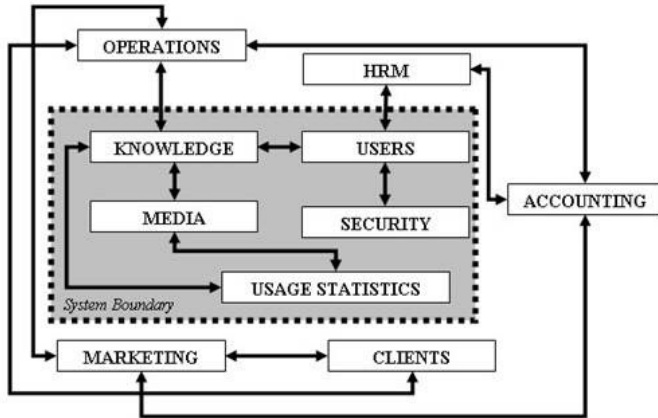


Causal Loops

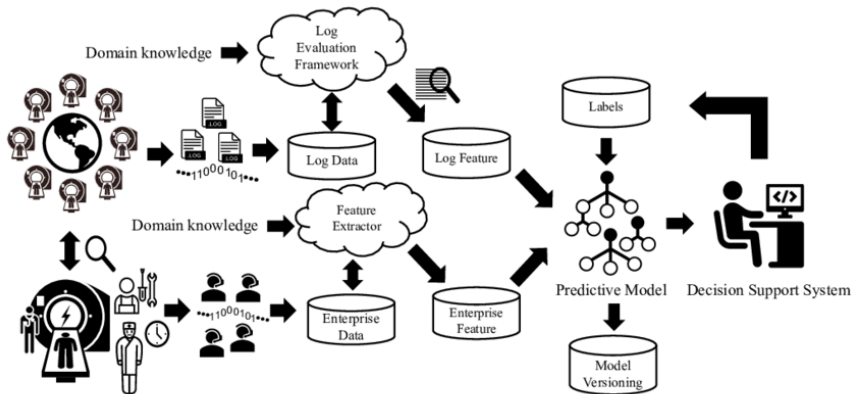
- A **Causal Loop** is a **diagram** that shows the **relationships** between different variables in a system.
- Causal loops are used to **analyze** and **understand** the **dynamics** of a system.
- Causal loops can be **positive** or **negative**.



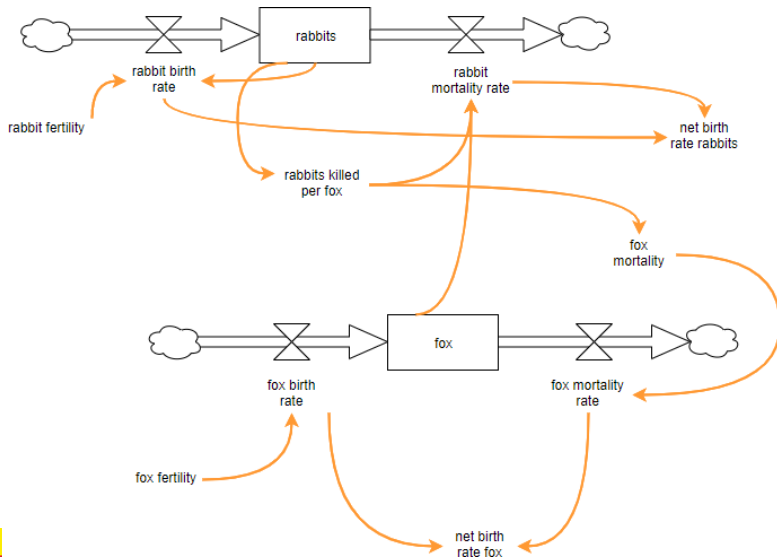
System Schema Example: Company Structure



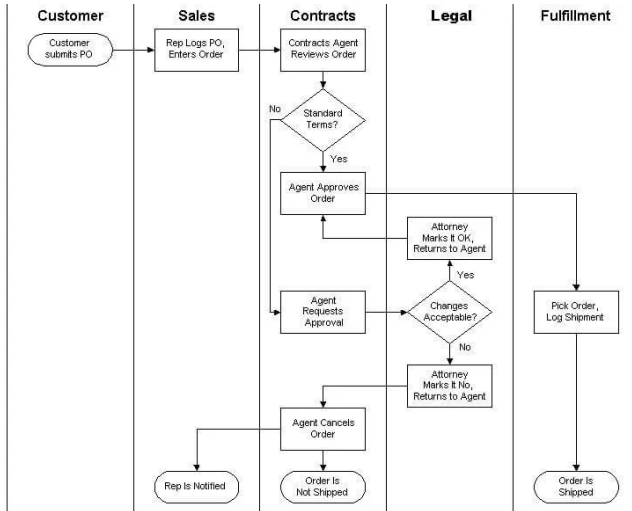
System Schema Example: Processing Pipeline



Stock and Flow Diagram



Business Process Model and Notation (BPMN)



Outline

1 Systems Thinking

2 Reference Architectures



Design Before Code

- **Design** should come **before coding**.
- Jumping into *code without a plan* leads to **confusion** and **rework**.
- **Good design** **clarifies** the problem and **guides** the solution.



Understanding the Requirements

- **Requirements** must be well understood before design.
- Ask questions, clarify ambiguities, and document all requirements.
- Requirements define the scope and direction of the design.

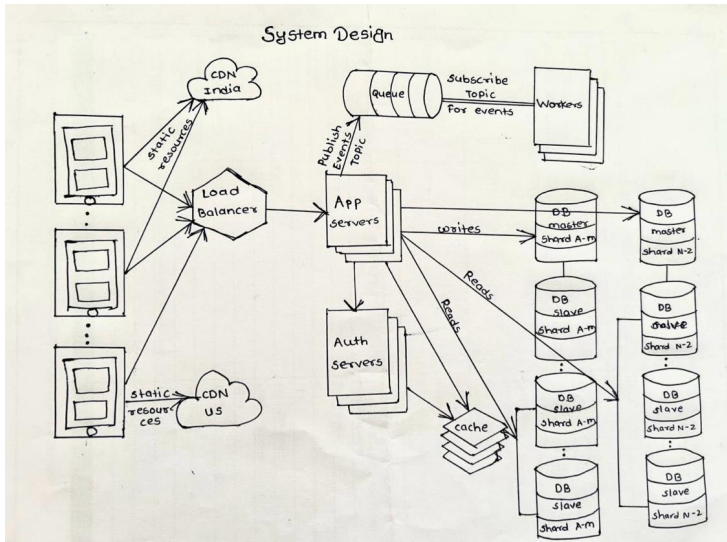


Design Based on the Problem

- **Design** should be *driven by the problem*, not by technology.
- Focus on what *needs* to be solved, *not just* how to implement it.
- Use the **problem statement** to identify *key objects* and their *relationships*.



Systems Design applied to Software Architectures



Conceptual Design and Technical Design

- **Conceptual Design:** What the **system should do**, using high-level models.
- **Technical Design:** How the **system will be implemented**, using detailed diagrams and specifications.
- Both are **essential** for a **successful** software project.



Conceptual Design and Technical Design

- **Conceptual Design:** What the **system should do**, using high-level models.
- **Technical Design:** How the **system will be implemented**, using detailed diagrams and specifications.
- Both are **essential** for a **successful** software project.



Conceptual Design and Technical Design

- **Conceptual Design:** What the **system should do**, using high-level models.
- **Technical Design:** How the **system will be implemented**, using detailed diagrams and specifications.
- Both are **essential** for a **successful** software project.



What is a System Architecture?

- A **system architecture** is the *structure* of a system that *defines* its **components**, **interactions**, and **relationships**.
- A **system architecture** is the *blueprint* of a system that *guides* its **development** and **implementation**.
- A **system architecture** is the foundation of a **system** that **ensures** that it **meets** the **needs** of its **users**.



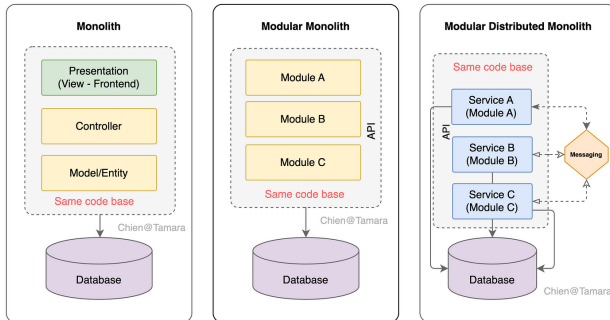
Types of System Architectures

- There are *several types* of **system architectures** that are *used* in systems development.
- They include **monolithic**, **client-server**, **peer-to-peer**, and **distributed architectures**.
- Each *type* of **architecture** has its own **advantages** and **disadvantages** that *depend* on the specific requirements of the **system**.



Monolithic System Architecture

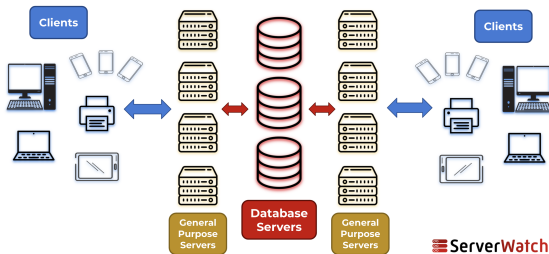
- A **monolithic system architecture** is a **single-tier architecture** that consists of a **single unit** that performs all the functions of the **system**.
- It is **simple**, **easy** to **develop**, and **maintain**, but it is **not scalable** and **flexible**. It is typically used for **small systems** that do not require **high performance** or **reliability**.



Client-Server System Architecture

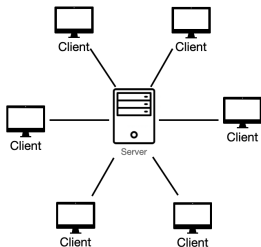
- A **client-server system architecture** is a **two-tier architecture** that **consists** of a **client** and a **server** that **communicate** with each other over a **network**.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is used for **medium** to **large systems** that require **high performance** and **reliability**.

The Client-Server Model

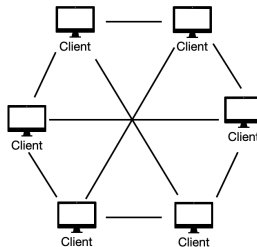


Peer-to-Peer System Architecture

- A **peer-to-peer system architecture** is a **two-tier architecture** that **consists** of a **network** of **peers** that **communicate** with each other directly.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is used for **medium** to **large systems** that require **high performance** and **reliability**.



Client Server Architecture

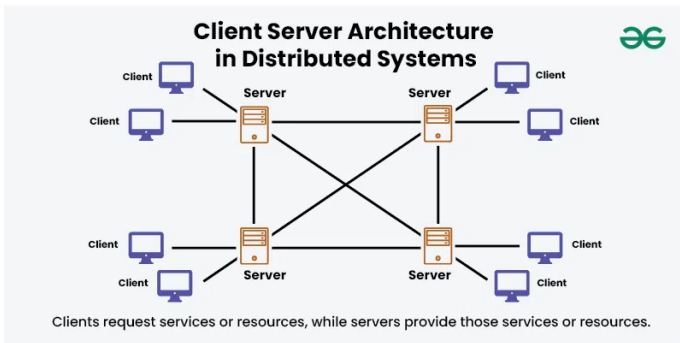


P2P Architecture



Distributed System Architecture

- A **distributed system architecture** is a **multi-tier architecture** that **consists** of a **network** of **nodes** that **communicate** with each other over a **network**.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is **used** for **large systems** that **require high performance** and **reliability**.

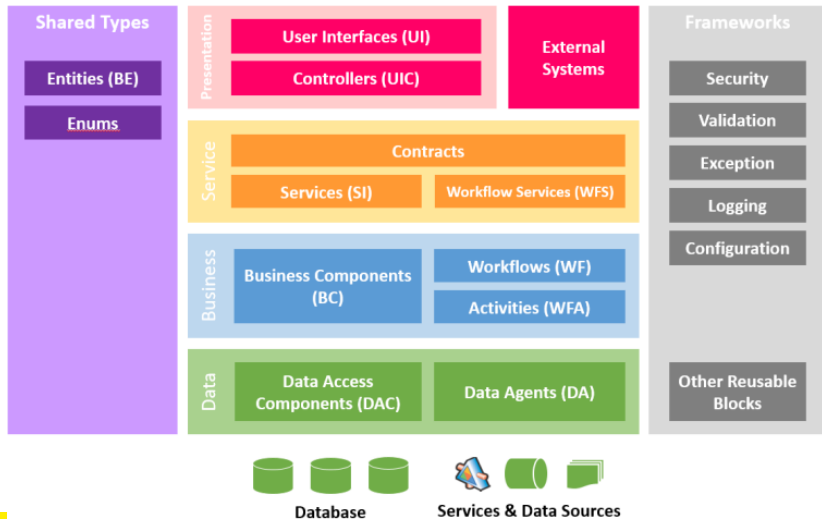


Software Architectures

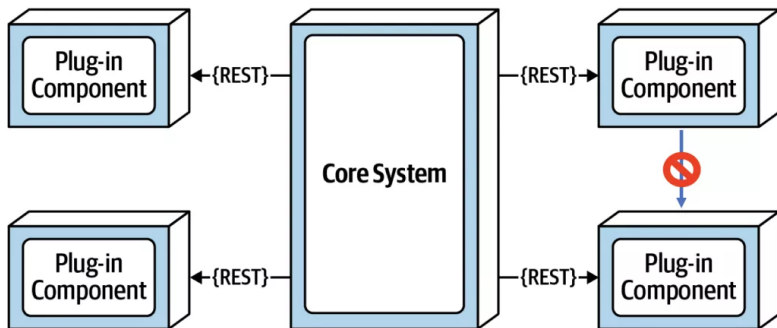
- A **software architecture** is a **high-level structure** of a **software system** that defines its **components**, **interactions**, and **relationships**.
- A **software architecture** is the *blueprint* of a **software system** that guides its **development** and **implementation**.
- A **software architecture** is the foundation of a **software system** that ensures it meets the needs of its users.



Layered Architecture Pattern

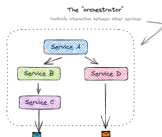


Microkernel Architecture Pattern



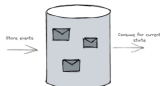
Event-based Architecture Pattern

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool



Process Manager

Central unit that orchestrates communication
Process a workflow in a single unit



Event sourcing

Store events and consume for state
Build current state using events



Event streaming

Stream of events usually used for real-time processing

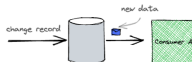
Inside event-driven architectures

What patterns will you come across when building event-driven architectures?



Point-to-point messaging (pull)

Messages put onto queue by sender
Receiver consumes messages from queue



Change data capture

Listening to events from changes in data
Consume events directly from DB

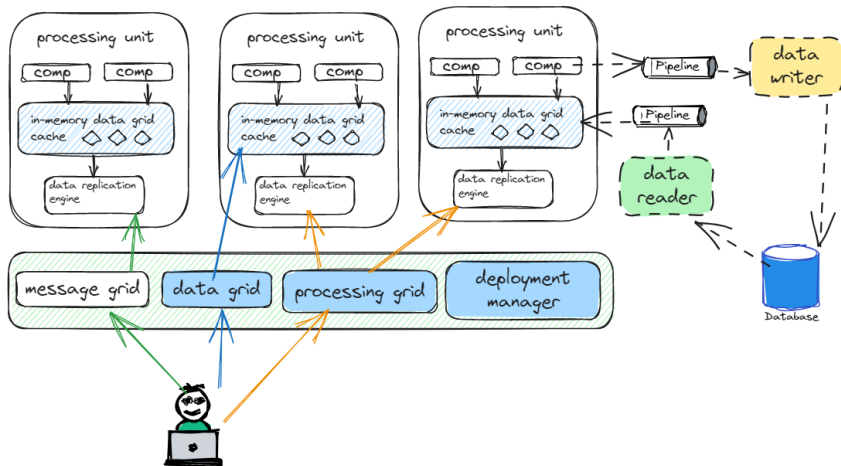


Pub/Sub (push)

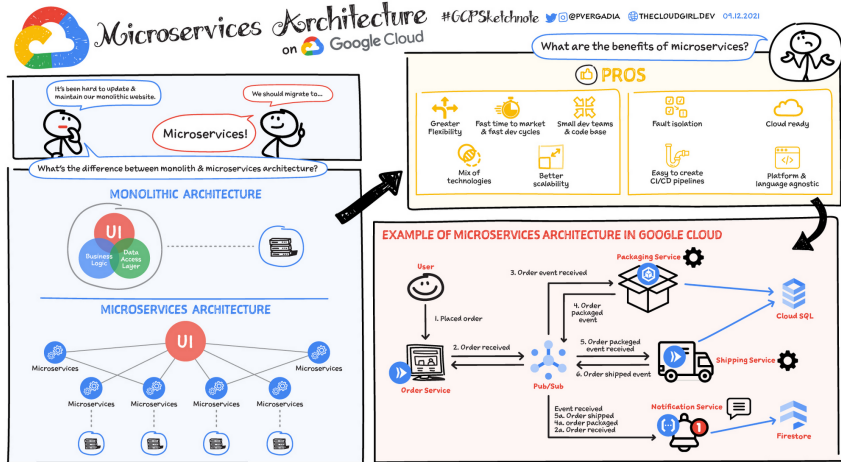
Publish messages/events to many subscribers
Each subscriber gets copy of event to process



Space-based Architecture Pattern

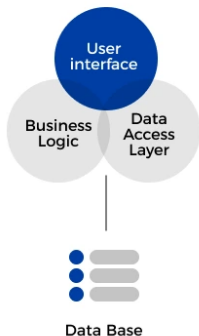


Microservices Architecture Pattern

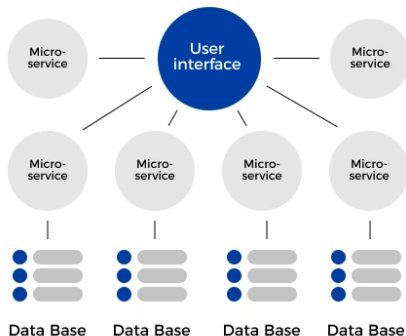


Monolithic vs. Microservices Architectures

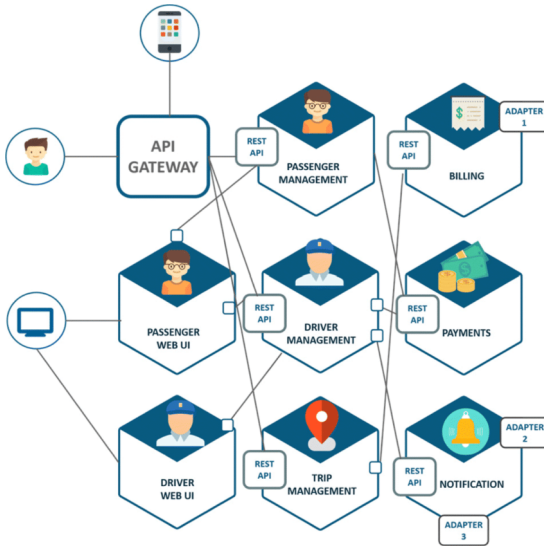
MONOLITHIC ARCHITECTURE



MICROSERVICE ARCHITECTURE



Case Study: Uber Microservices



Source: Kappagantula 2018



Outline

1 Systems Thinking

2 Reference Architectures



Thanks!

Questions?



Repo: www.github.com/EngAndres/ud-public/tree/main/courses/software_engineering_seminar

